

Law Database Enrichment Project

Technical & Executive Report

Arabic Legislation Database — Data Quality Initiative

Prepared by

Data Engineering Team

Date

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Status

Completed — 99.8% Resolution Rate

Records Processed

3,542

Classification

Internal

1. Executive Summary

In plain language: Our law database had thousands of records with missing information. We built an intelligent system that automatically found and filled in that missing information by cross-referencing three separate data sources. The result: 99.8% of all records are now complete — only 8 records out of 3,542 could not be resolved, and those were individually investigated.

This report documents the full process — from problem identification through technical design, execution, results, and database update procedures — in a way that is clear to both technical and non-technical readers.

Key Metric	Result
Total law records in database	3,542
Records successfully matched & enriched	3,534 (99.8%)
Individual data fields filled in	2,226
Database rows to be updated	1,616
Records needing manual review	8 (0.2%)
Data sources cross-referenced	3 files
Matching techniques applied	5 layers

2. The Problem

2.1 What Was Missing

The primary legislation database contained 3,542 law records, many of which had critical fields left empty. These gaps produced incomplete search results, unreliable legal references, and reduced trust in the system as an authoritative source.

Field Name	Empty Records	% Incomplete	Impact
Active_Date (law effective date)	1,439	40.6%	Cannot determine when a law took effect
Replaced_For (superseded laws)	3,516	99.3%	Cannot trace legal lineage
Magazine_Number (gazette ref.)	140	4.0%	Cannot cross-reference official gazette
Magazine_Date (gazette date)	151	4.3%	Cannot verify publication date
Magazine_Page_Number	64	1.8%	Cannot locate exact gazette page

2.2 Root Cause

The gaps existed because the database was originally populated from multiple independent sources that used different naming conventions, different law number systems, and inconsistent Arabic text formatting. No automated link existed between these sources.

3. Our Approach

3.1 Data Sources Used

File	Description	Records
law_merged_output_V2.xlsx	Primary LOB database — the file we enriched	3,542
cleaned_v03_merged_updated.xlsx	Enhanced legislation dataset with additional metadata	4,701
amendment_4_5_2026.csv	Amendments register linked via LobID foreign key	5,089

3.2 Arabic Text Normalization

Before any matching, all law names were standardized to remove inconsistencies that exist across Arabic legal databases:

- **Alef variants:** آ / إ / أ all converted to ا
- **Teh Marbuta:** ة converted to ه
- **Yeh variants:** ي converted to ي
- **Tatweel removed:** elongation marks (ـ) stripped
- **Diacritics removed:** short vowel marks stripped
- **Punctuation normalized:** parentheses and commas handled consistently

Example: "قانون معدل لقانون الهيئة" (with Tatweel marks) and "قانون معدل لقانون الهيئة" (without) become identical after normalization, allowing them to match correctly.

3.3 The 5-Layer Matching Pipeline

We built a 5-layer pipeline where each layer handles a different category of match. Records pass through layers in order — the first successful match wins.

Layer	Name	Logic	Records Resolved
1	Exact 3-Key Match	Matches on Law Number + Year + Magazine Number together. Zero ambiguity.	1,584 (44.7%)
2	TF-IDF Cosine Similarity	Converts law names to mathematical vectors using character patterns, then measures similarity. Handles word-order differences and	147 (4.2%)

Layer	Name	Logic	Records Resolved
		extra words.	
3	Substring Containment	Checks if a short name appears inside a longer formal title. Safety net layer.	0 (safety net)
4	LobID Foreign Key	Uses LobID as a direct database link to the amendments register. No name matching needed — a direct pointer between records.	1,651 (46.6%)
5	Amendment Reference Extraction	For laws titled "Amendment to Law X", extracts "X" and searches for the original law. Uses Year as a tiebreaker: an amendment cannot predate its original law.	152 (4.3%)

4. Results

4.1 Matching Performance — Layer by Layer

Layer	Records Matched	Running Total	Running %
Layer 1 — Exact 3-Key	1,584	1,584	44.7%
Layer 2 — TF-IDF Cosine	147	1,731	48.9%
Layer 3 — Substring	0	1,731	48.9%
Layer 4 — LobID Foreign Key	1,651	3,382	95.5%
Layer 5 — Amendment Reference	152	3,534	99.8%
Unresolved	8	8	0.2%

4.2 Data Fields Enriched

Field	Was Empty	Now Filled	Gaps Closed
Active_Date	1,439	1,197	83.2%
Replaced_For	3,516	775	22.0%
Magazine_Number	140	123	87.9%
Magazine_Date	151	74	49.0%
Magazine_Page_Number	64	57	89.1%
TOTAL	5,310 gaps	2,226 filled	41.9% overall reduction

4.3 The 8 Unresolved Records

These are not algorithm failures — they are source data quality issues:

Record	Root Cause	Recommended Action
"قانون معدل لقانون العمل" x3	Same law entered three times	De-duplicate in source; keep one

Record	Root Cause	Recommended Action
(1996)	with different numbers — duplicate entries	canonical record
"قانون معدل لقانون السير" x2 (2008)	Two amendments in same year with identical gazette numbers — genuinely ambiguous	Verify against original gazette issue
"قانون الهيئة الوطنية لازالة الالغام"	Absent from all three digital sources	Source from official gazette archive
"قانون محاكم العشائر" (1933)	Pre-digital era — no digital record exists	Scan physical gazette from 1933
"نموذج" (1900)	Means "template/form" — a data entry placeholder, not a real law	Delete or mark as invalid

5. Data Safety & Quality Guarantees

5.1 Non-Destructive by Design

IMPORTANT: The pipeline NEVER overwrites existing data. It only fills in fields that were previously NULL (empty). If a record already has a Magazine_Number, it is left completely untouched.

5.2 Full Audit Trail

Every change is documented in law_changes_audit.xlsx:

- **Changes Log:** every cell changed — old value (NULL) → new value, which layer matched it, confidence score
- **Summary:** total counts by column and by matching technique
- **Unresolved:** the 8 records flagged for manual review with analysis
- **Unchanged:** all records where no changes were made

5.3 Approval-Gated Database Update

The database is NOT updated automatically. A human must approve first:

Step	Action	Responsible
1	Review law_changes_audit.xlsx and spot-check matched records	Domain Expert
2	Confirm disposition of 8 unresolved rows	Legal Team
3	Run law_update.sql (wrapped in BEGIN TRANSACTION)	DBA
4	Verify 1,616 rows updated in database	DBA
5	COMMIT if correct — ROLLBACK if any issue found	DBA

6. Frequently Asked Questions

Q: How do we know the matches are correct?

Each match is tagged with its technique. Layer 1 and Layer 4 are mathematically certain (exact key or direct foreign key). Layers 2, 3, and 5 include a confidence score from 0.0 to 1.0. The Changes Log lets any reviewer trace exactly why each cell was filled in, and by which source.

Q: What if a match turns out to be wrong?

The SQL update runs inside a BEGIN TRANSACTION block. The entire operation can be reversed instantly with a single ROLLBACK command — restoring the database to its exact previous state with zero data loss.

Q: Why not use a simple name search?

Arabic law names are highly inconsistent: the same law may appear as "قانون الشركات" in one file and "قانون رقم 22 لسنة 1997 (قانون الشركات لسنة 1997) وتعديلاته" in another. A simple text search would have failed on over 55% of records. The 5-layer approach was necessary to handle real-world data inconsistency.

Q: Can this be re-run when new data arrives?

Yes. The pipeline is a standalone Python script. Updating the file paths and re-running produces a fresh enriched file, audit workbook, and SQL script. No code changes needed for routine updates.

Q: What about the 0.2% that could not be resolved?

The 8 unresolved records were individually analyzed (see Section 4.3). Five are duplicate or placeholder entries in the source data itself, two require gazette archive research, and one is genuinely absent from all digital sources. These represent the irreducible limit of automated matching — not algorithm failures.

7. Deliverables

File	Purpose	Audience
law_merged_output_V2_updated.xlsx	Clean updated database — identical structure to original, gaps filled	Supervisor / Database Team
Law_Enrichment_Report.docx	This report	All stakeholders
law_changes_audit.xlsx	Full audit trail of every change made	Domain Expert / DBA
law_update.sql	1,616 SQL UPDATE statements ready for database execution	DBA
law_matching_v5.py	5-layer matching pipeline — main engine	Data Engineer
law_db_writeback.py	Generates audit workbook and SQL from enriched output	Data Engineer
law_clean_export.py	Generates clean updated Excel for supervisor	Data Engineer
mismatch_report_v5.xlsx	8 unresolved records with individual analysis	Legal Team

8. Conclusion

Starting from a database with over 5,000 empty fields and no reliable way to automatically link records across files, we designed and delivered a 5-layer intelligent matching pipeline that achieved a 99.8% resolution rate — filling 2,226 individual data fields across 1,616 database records. The remaining 8 records (0.2%) were individually analyzed and confirmed to be source data quality issues, not algorithm failures. The entire process is documented, auditable, reversible, and repeatable.

The pipeline is production-ready and can be re-run whenever new source data is available, ensuring the database stays current with minimal manual effort.

— End of Report —